# ME-420


## Group 1

**Reconfigurable Robotics Laboratory**
**EPFL, Switzerland**

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# ScrewSorter

# AGENDA

1/ Motivation and State of the art

2/ Design progression

3/ Final design : mechanics, electronics and control loop

4/ Proof of concept : Performance evaluation

5/ Sustainability evaluation

6/ Future development

# PROBLEM STATEMENT

Many small to medium-sized workshops face the problem of unsorted screws, which leads to:

Screws being discarded instead of reused, causing **waste, energy emissions + ressource consumption + efficiency**

- **Waste**: 10 kg/year (~3,000–5,000 screws)
- **Impact**: 1 kg of screws = 7.67 kg $CO_2e$

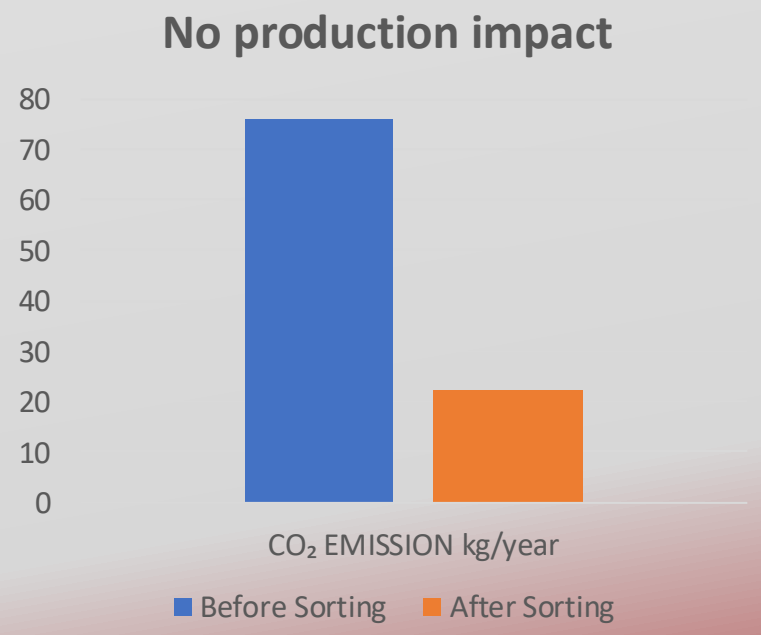
and a Water deprivation potential (WDP) of 2460 liters (crade-to-gate)[1]

**=> 76.7 kg $CO_2e$/workshops**

# MOTIVATION

Being able to reuse most screws of the workshops quickly and efficiently at low cost and low energy consumption

70% reuse → ~7 kg screws saved → ~54 kg $CO_2$e avoided / workshop / year
(= driving a car for 300 km Lyon – Geneva)
→ 17220 liters of water preserved / workshop / year



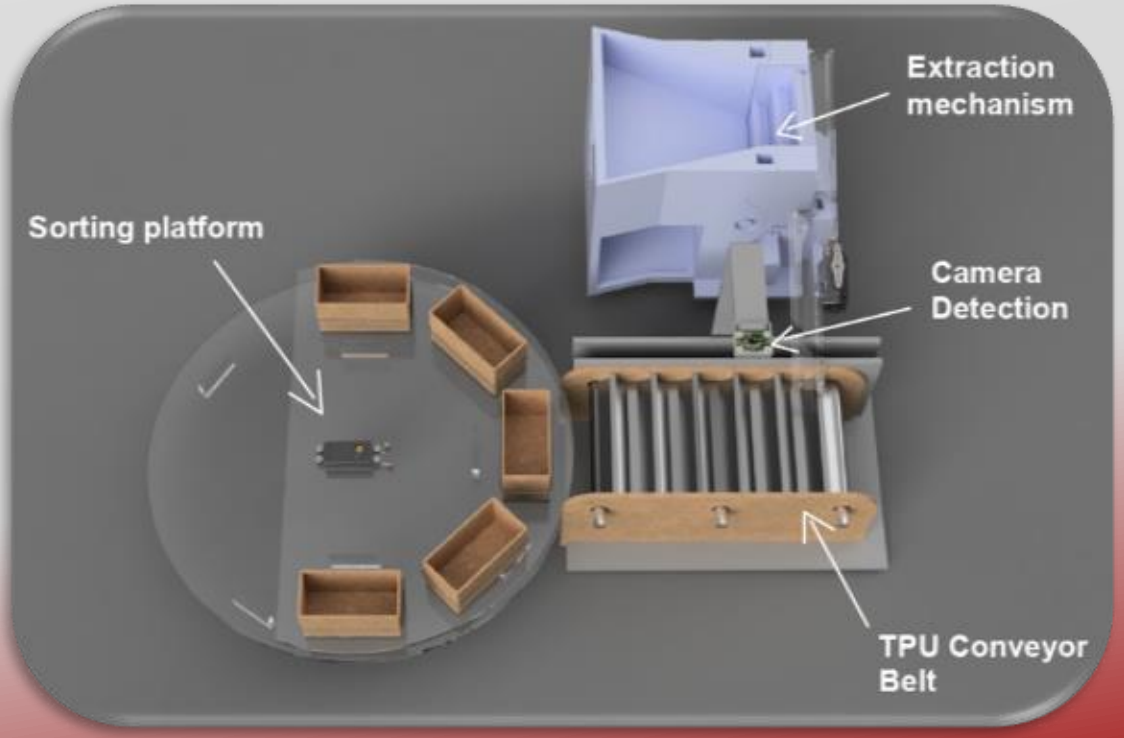

**Scope of Impact:**

- **Human:** Saves labor hours
- **Societal:** Promotes circular economy in workshops
- **Environmental:** Less steel demand → less mining → less $CO_2$ and less water used and/or polluted

# OUR IDEA

Develop an intelligent automated sorting system that requires only one action from the user: placing the screws into an initial bowl. The machine then performs a series of fully automatic steps:

1. **Screw Isolation**
2. **Camera-Based Diameter & Length Measurement**
3. **Automated Sorting Into Bins**



# STATE OF ART

**Manual Sorting**

- manual sorting of size : **4 screws / min**
- manual sorting of length : **8 screws / min**
- automatic sorting of length : **10 screws / min**

**Industrial sorting machines:** [2]

- Expensive
- Consumes lots of energy
- Too big for small workshops.

# NOVELTY

- Small-scale and energy-efficient system
- Designed for small workshops and makerspaces
- Modular, repairable, and can be built with recyclable materials
- Automatically sorts multiple diameters and lengths **without any hardware modifications**

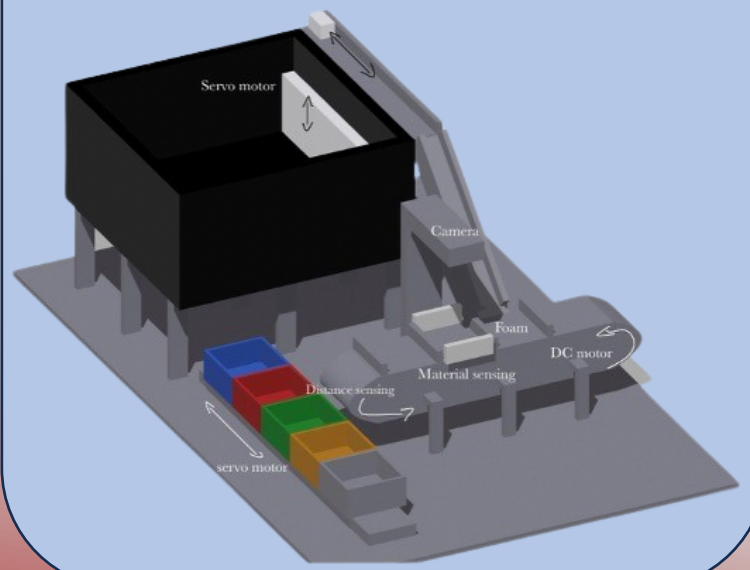
**=> Reduces operator workload, saves time, and lowers $CO_2$ emissions**

# ENGINEERING SPECIFICATION

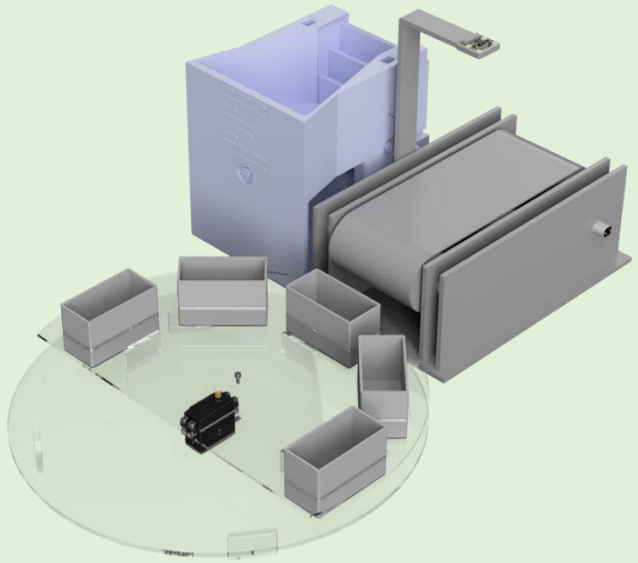
| | |
|---|---|
| **A/ Number of screw sorted :**<br>-> Be faster than doing it manually | **15 screws per minute** |
| **B/ Accuracy of sorting:**<br>-> Be as close as possible to manual-sorting accuracy | **90%** |
| **C/ Energy consumption:**<br>-> Be less energy-intensive than larger machines of big industry | **max 70W instantaneously** |
| **D/ Volume:**<br>-> Be usable and manageable in a medium-sized workshop | **700 mm x 1000 mm x 500 mm** |
| **E/ Screws Range :**<br>-> Sort as much as possible different screws | **M3-M8; L20-L80** |
| **F/ Material sorting accuracy:**<br>-> Additional functionality to sort screws by material (magnetic) | **90%** |

# DESIGN PROGRESSION

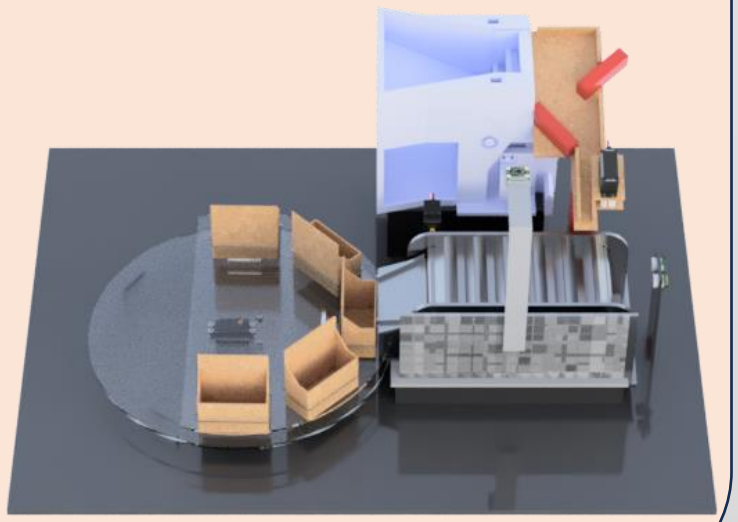
## Design ideas




## Prototyping & Iterations



## Final Prototype

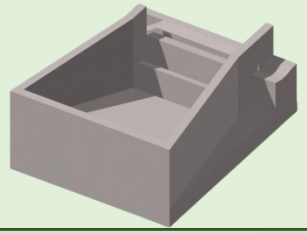


# STEPS OF DESIGN

1. One-by-one screw pickup from bulk

**Vibrating bowl**

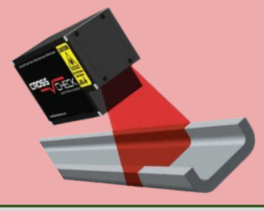
**Back and forth motion**



**Gripper**

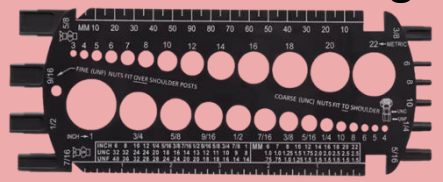
2. Diameter screw sorting

**Laser measurement**



**Camera inspection**

**Mechanical sorting**

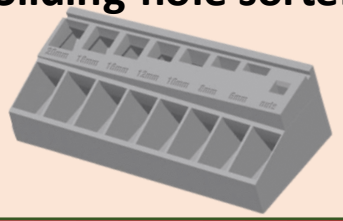


3. Length screw sorting

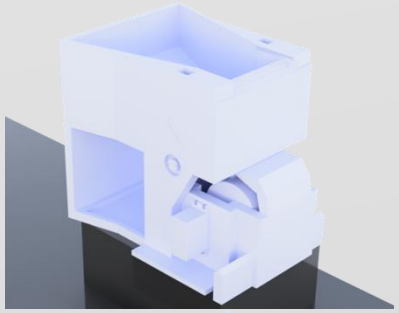
**Overhead camera**

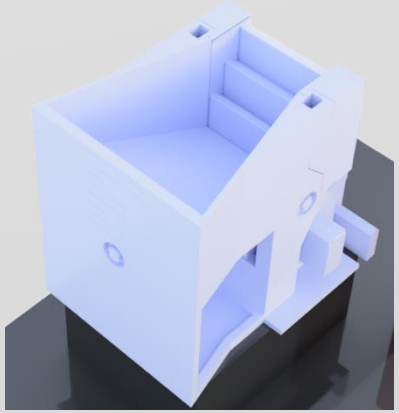
**Laser measurement**

**Sliding-hole sorter**



# MOTION : isolate mechanism



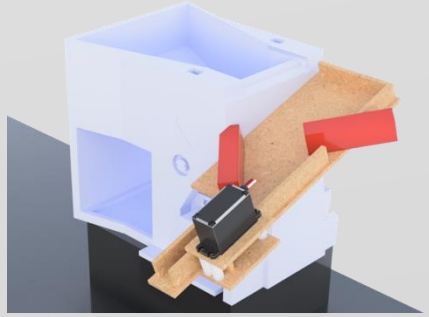


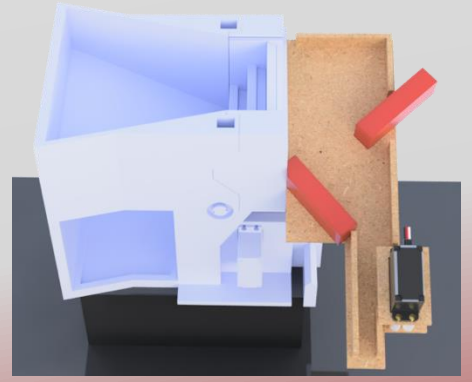
Lifter System:
- Inspired from state of the art
- Actuated by continuous rotation servo + excentric cylinder combination
- Staircase compatible for screws up to M6 and 70mm length

Problem:
Multiple smaller screws picked up at once
--> Not suitable for camera

Solution:
Lifter + Single Screw Isolation Ramp Combination

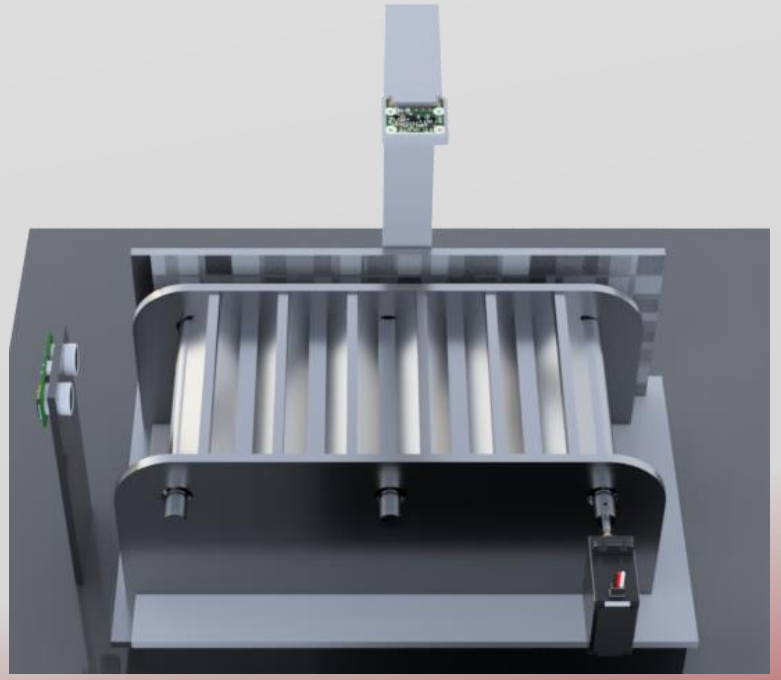
- Addition of ramp for screw isolation
- Connected to Lifter System
- Pushes screws similar to a pinball system





# MOTION: Conveyor

- **Objective:** The controlled transportation of individual screws from the extraction subsystem to the sorting subsystem, while providing real-time data acquisition through an integrated camera and copper coil.
- **Main constraints for the design:**
  - Special valley-shaped sections ensure proper screw alignment for accurate camera detection.
  - Partition walls between belt sections facilitate the implementation of a closed-loop control strategy, enabling precise detection and correction of belt slipping and jumping events.

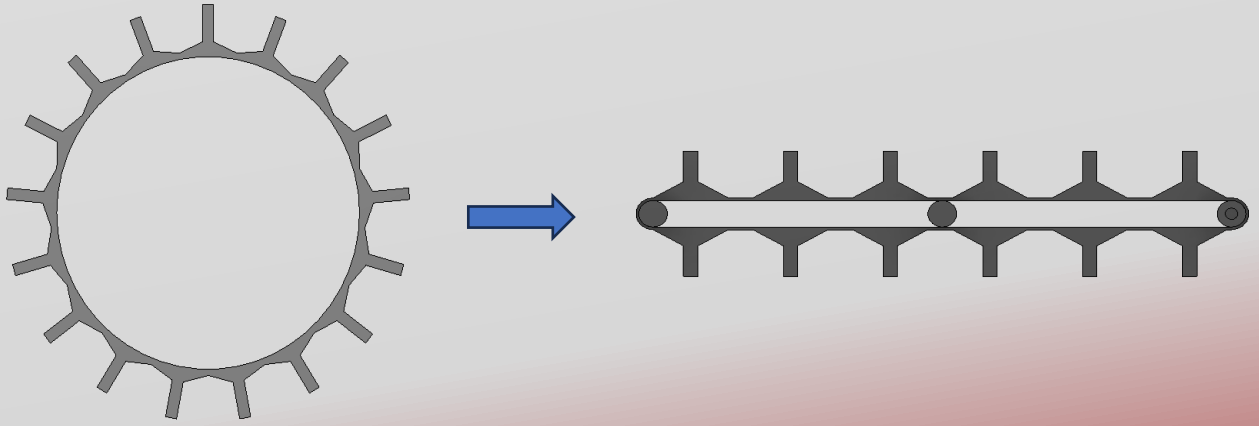


- For the overall conveyor design, we adopted a traditional three-roller configuration, with one roller driven by a continuous servo motor.
- The rollers were machined from aluminum to provide a lightweight and sufficient structural resistance capable of withstanding the belt tension.

# MOTION: Conveyor

- A custom TPU belt was designed and 3D-printed to meet the geometric and functional constraints of the system.

- Rubber bands were cut and glued onto the aluminum rollers to increase the friction coefficient between the belt and rollers, thereby increasing the slipping threshold and preventing belt slip during actuation.

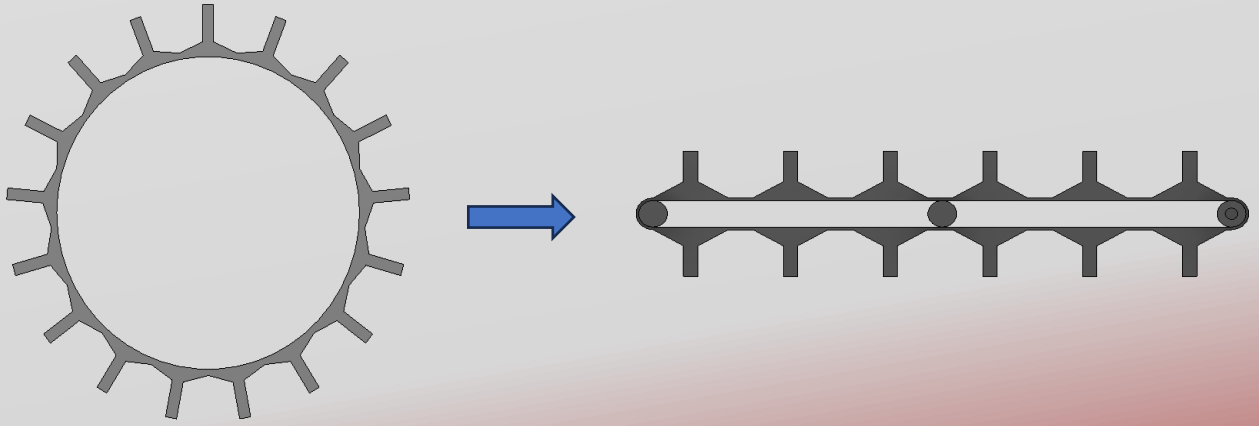


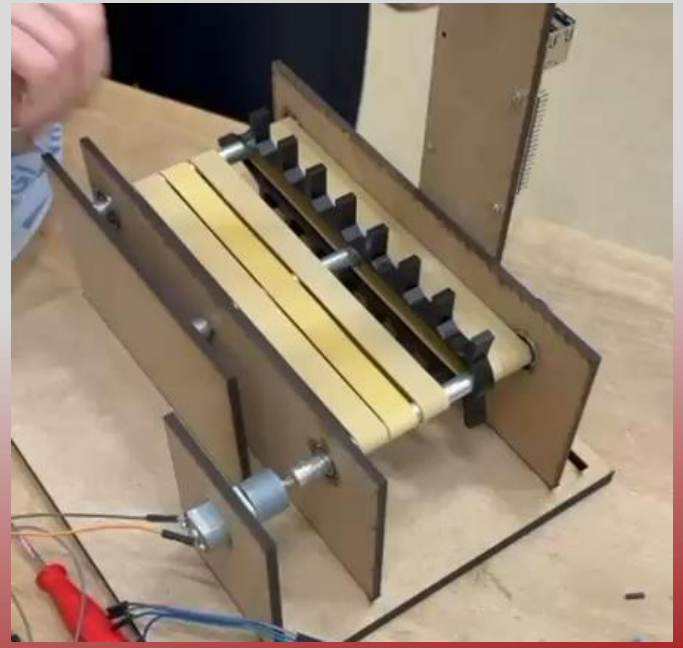
```
=== TPU BELT MECHANICAL ANALYSIS ===

Initial perimeter P1:            0.424115 m
Installed perimeter P2:          0.439000 m
Belt strain epsilon:             0.0351 (3.51%)
Cross-sectional area A:          4.983000e-04 m^2
Belt tension T:                  209.864 N
Normal force N:                  659.306 N
Max friction force F_friction:   791.167 N
Slip torque limit M_slip:        4.905 N·m
Required torque M_required:      2.800 N·m
Servo torque M_servo:            2.940 N·m

RESULT: Belt WILL NOT slip (2.800 < 4.905)
```

# MOTION: Conveyor

Because of the belt's complex behavior around small-radius rollers, the torque required to move the belt could not be calculated analytically. Multiple factors make the system too nonlinear for a closed-form solution.
We therefore determined the required drive torque experimentally and found that a value of about 2.8 N·m was needed for smooth rotation.

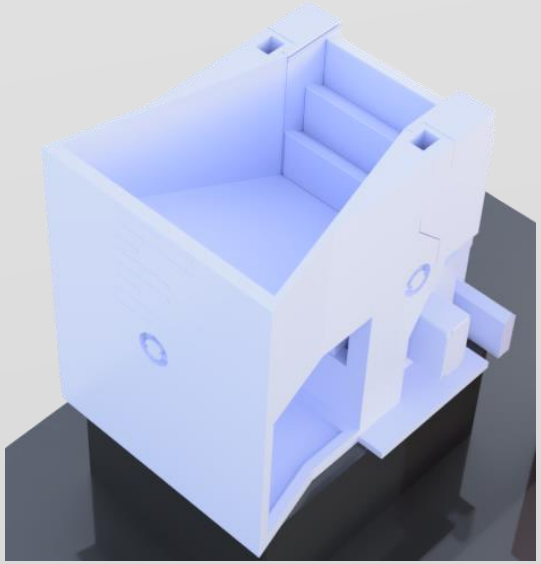
When the belt passes over a very small roller, it momentarily creates a significant reduction in contact surface and produces a jumping effect. This intermittent loss of contact causes the belt to shift and slip during actuation.



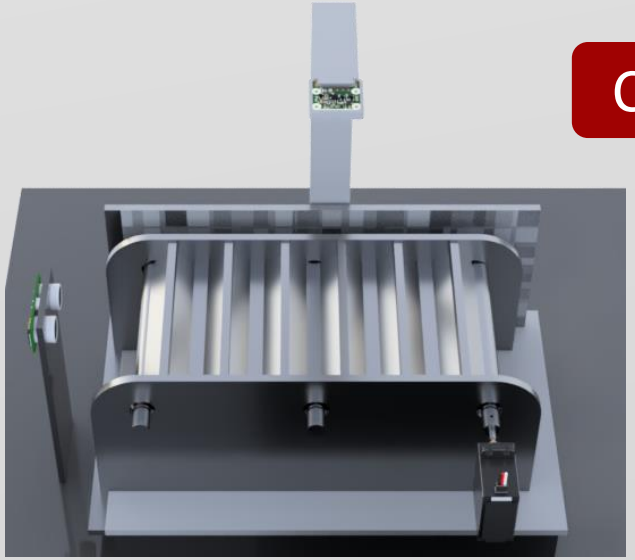


# FINAL DESIGN : Features

Lifter

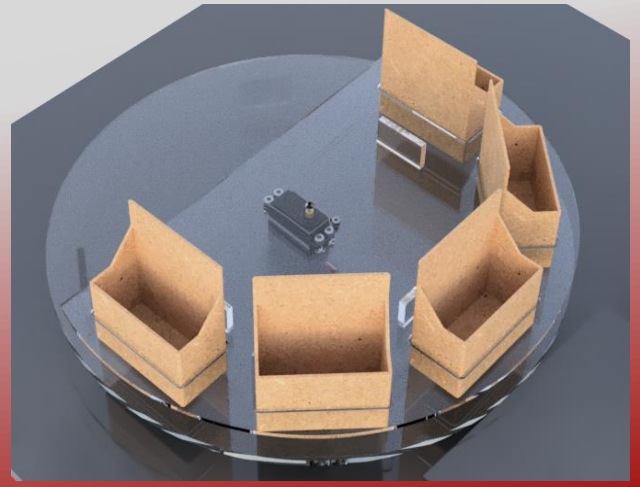


Conveyor



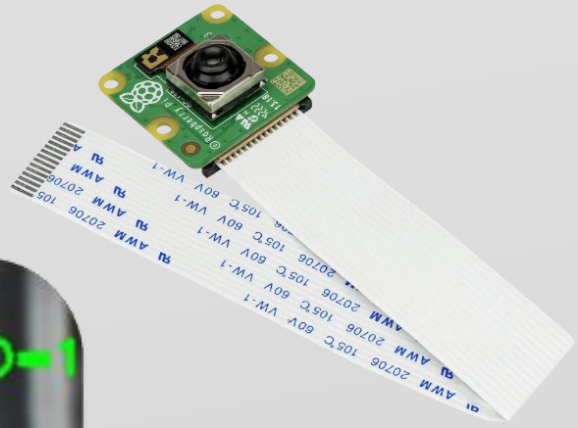
Screw Separator

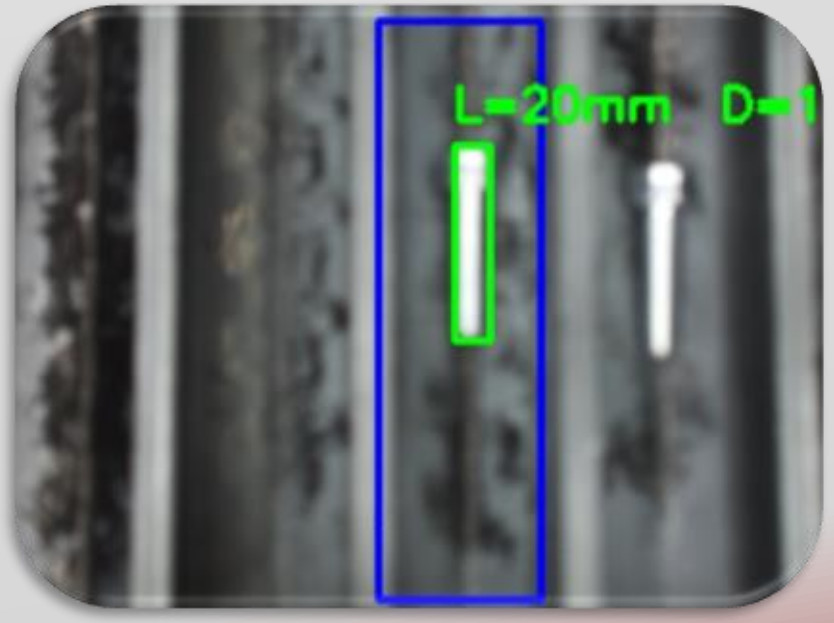
Bins



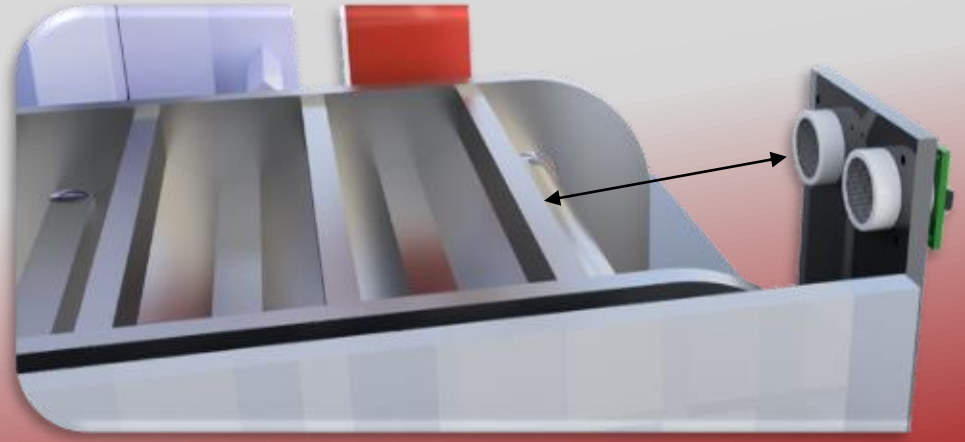
# FINAL DESIGN : Features

Camera






Distance sensor



# FINAL DESIGN : Electronics

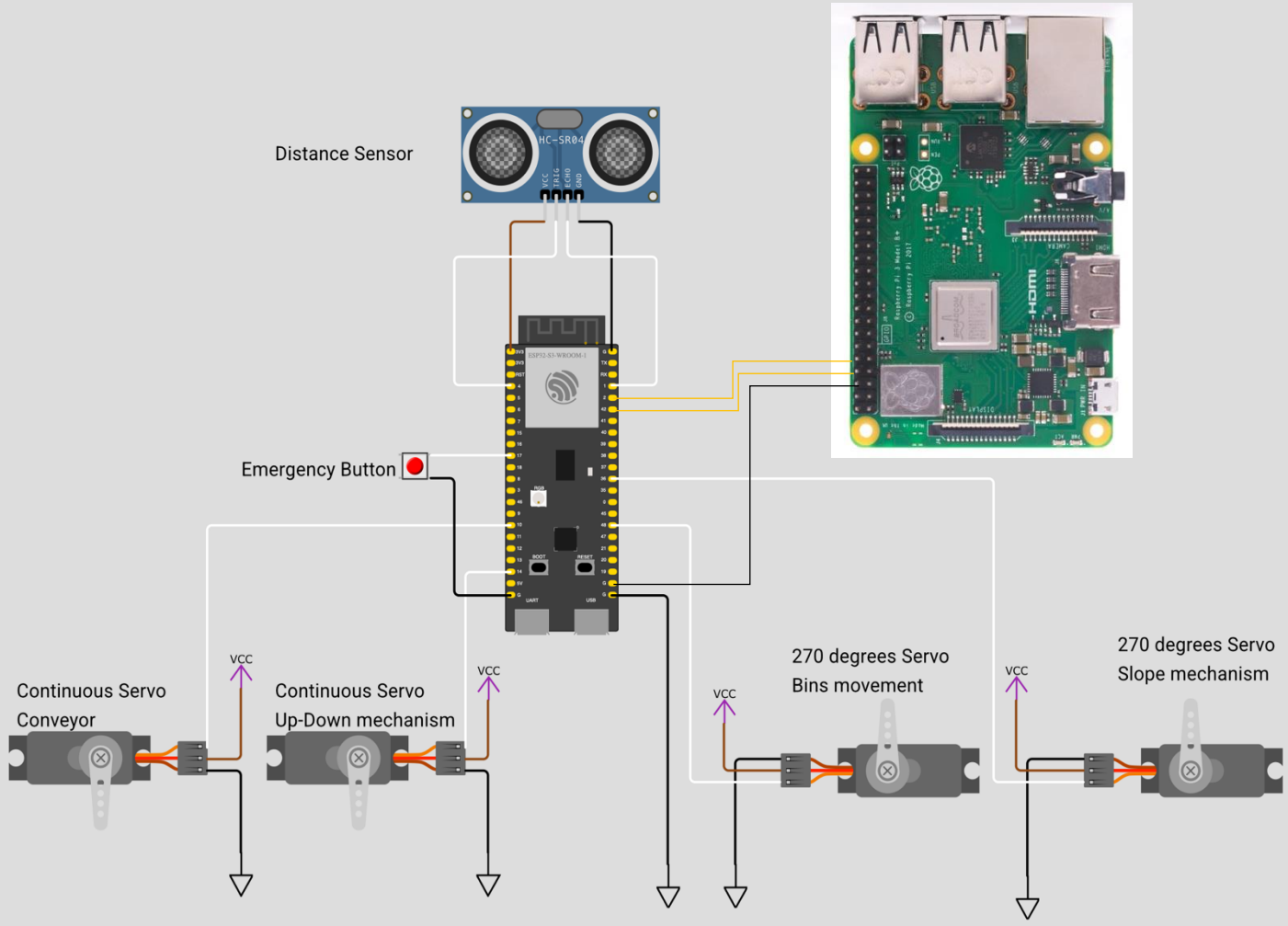
**Important :**

- Connect all the grounds together
- Not same ground for ESP and Raspberry PI

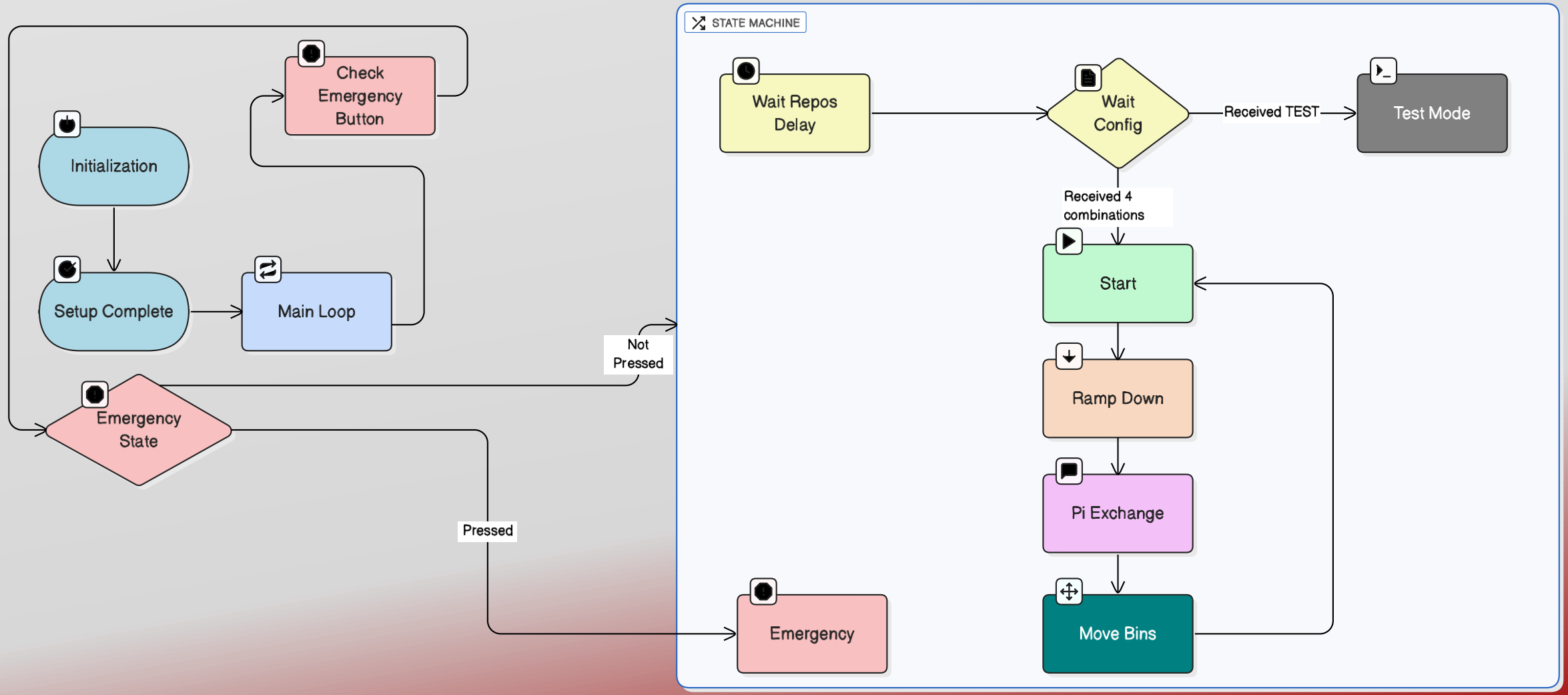
→ no UART communication

- Not same ground for ESP and the motors

→ The servos don't move as wanted



# SOFTWARE FLOWCHART



# USER INTERFACE

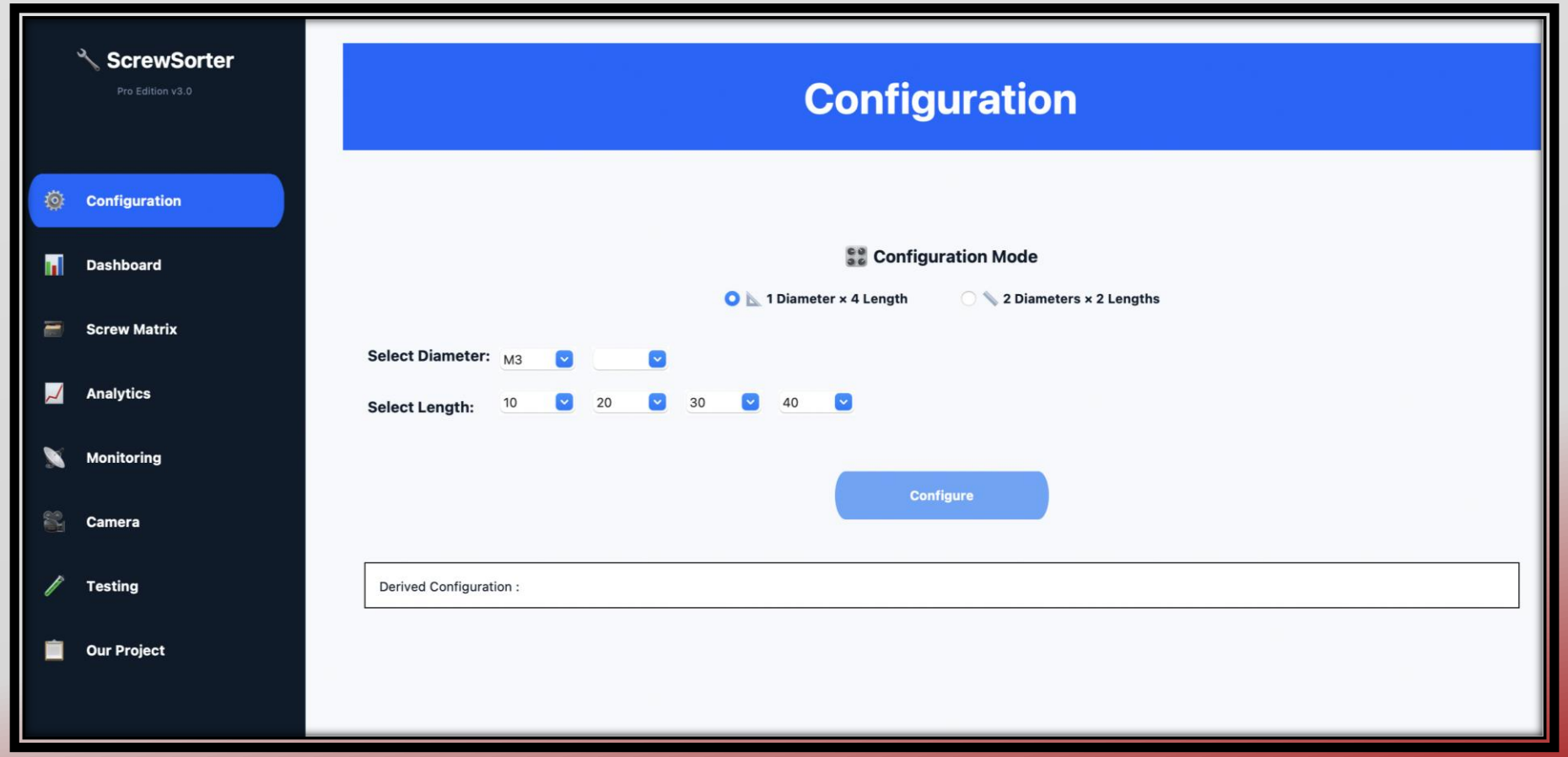
- **Configuration** :
The user can choose the screw combinations and send it to the ESP32

- **Dashboard** :
Live-time count of many screws have been sorted

- **Analytics** :
Live-time graphs of speed and electric power

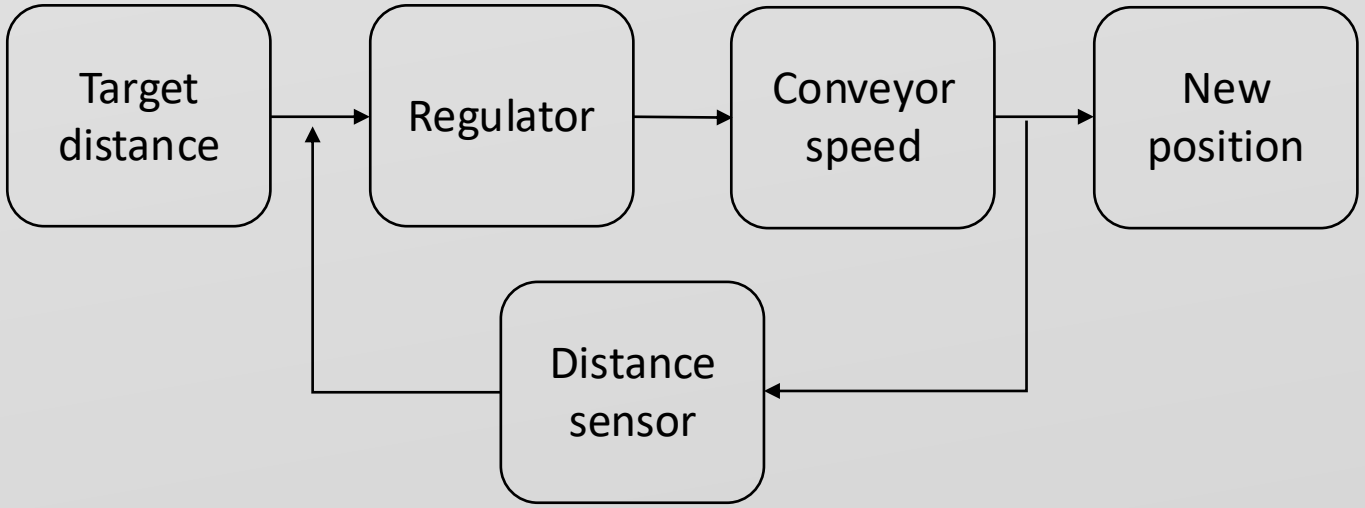
- **Monitoring** :
Messages sent by the ESP32

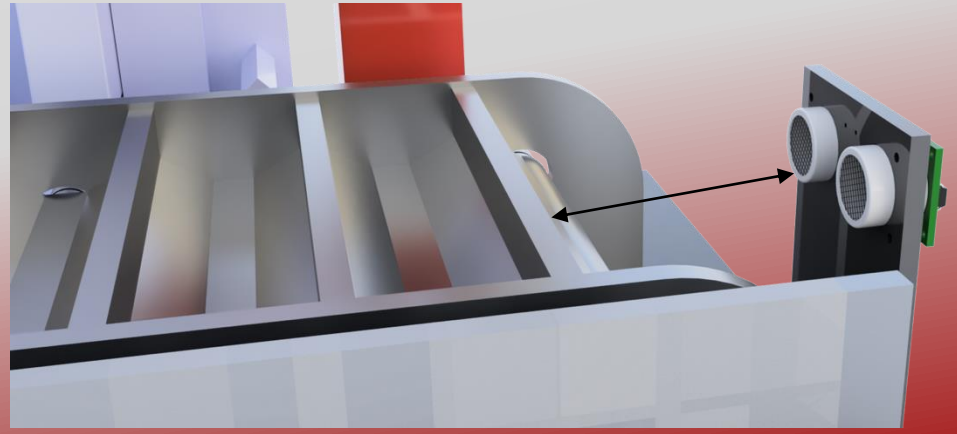
- **Testing** :
Give the user the possibility to test every motor one by one



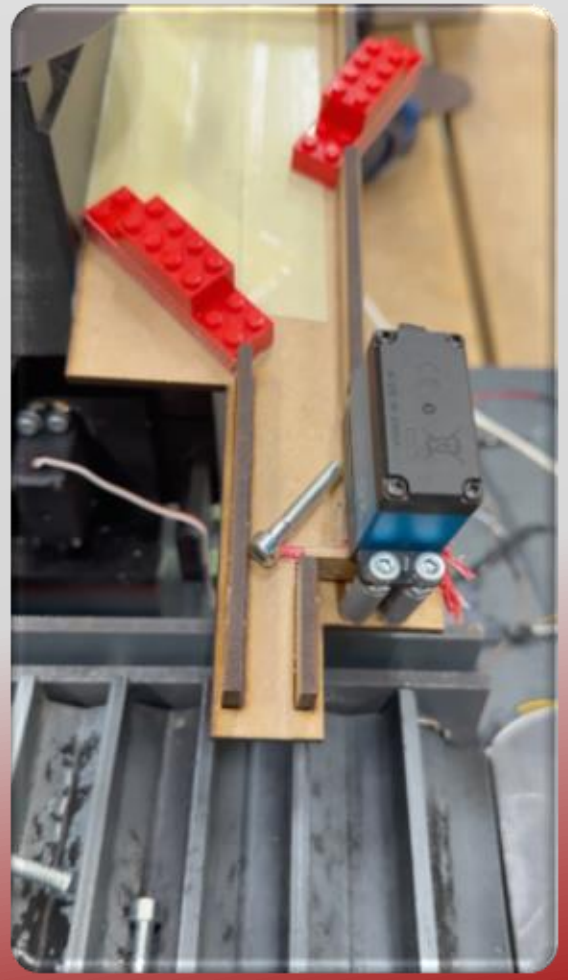
# Closed loop control

- Used to stop the screw just under the camera
- Target distance of 9.15 cm determined by placing the belt in a way that the screw is the camera's field on vision
- Proportional control to slow down the belt and stops it at the right place
- 750ms where the motor turns at constant speed to pass the arrival of the new wall before the controlled speed



# PROOF OF CONCEPT



# PROOF OF CONCEPT

1. The conveyor stops under the camera

2. Constant speed to avoid the new wall coming

3. Adjusted speed with the distance

**Importance of the closed loop :**

- As we can see, the distance sensor varies a lot.

→ We can't have a simple solution where the conveyor stop when a certain distance is measured

# PROOF OF CONCEPT

- Maximum power of 57W when the conveyor moves
- Peaks appear when the bins of the slope servo are active

# PERFORMANCE EVALUATION

| Criterion | Value expected | Value achieved |
|---|---|---|
| A/ Number of screw sorted | 15 screws/min | 12 screws/min |
| B/ Accuracy of sorting | 90% | 90% |
| C/ Energy consumption instantaneous | 70W max | 57 W max |
| D/ Volume | 700 mm x 1000 mm x 500 mm | 600 mm x 900 mm x 280 mm |
| E/ Screws range | M3-M8; L20-80 | M3-M6 ; L10-70 |
| F/ Material sorting accuracy | 90% | 70% |

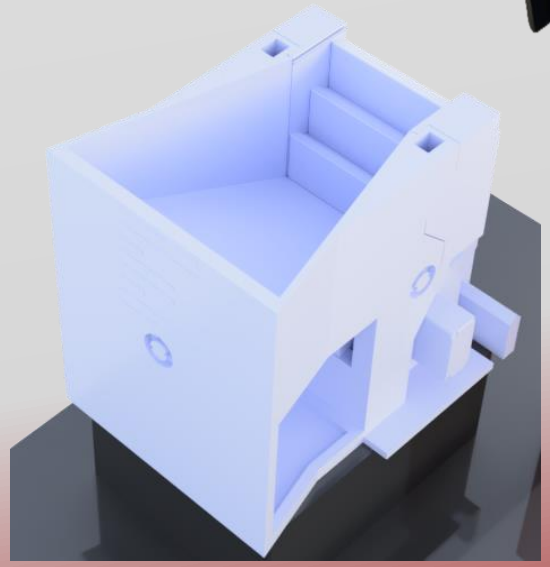
# COST/COMPONENTS ANALYSIS

| Components | Quantity | Price (CHF) |
|---|---|---|
| Conveyor motor | 1 | 36,30 |
| Lifter motor | 1 | 11,70 |
| Separator motor | 1 | 12,40 |
| Bins motor | 1 | 12,40 |

= 72,80 CHF

# COST/COMPONENTS ANALYSIS

| Components | Quantity | Price (CHF) |
|---|---|---|
| TPU belt | 1 | 30,53 |
| 3D printing Lifter | 1 | 16,54 |
| MdF laser cut | 4m^2 | 10,00 |
| Plexiglas laser cut | 1m^2 | 3,00 |
| Alu axes | 3 | 3,00 |



= 53,07 CHF

# COST/COMPONENTS ANALYSIS

| Components | Quantity | Price (CHF) |
|---|---|---|
| Power supply 5V 5A | 3 | 24,20 |
| Camera | 1 | 28,70 |
| Rasbery Pi | 1 | 35,70 |
| ESP32-S3 | 1 | 10,78 |
| Wires | 1 | 10,30 |
| Dominos | 1 | 6,50 |
| Esp/rasbery pie cables | 1 | 10,50 |

= 126,68 CHF

# COST/COMPONENTS ANALYSIS

**- Cost of the total prototyping :** **262,55 CHF**

= cost of all components, needed to build the final prototype of a single unit

**- Cost of final production cost (for 100 product) :** **187,59 CHF**

= cost to produce one unit within a batch of 100, which lowers the individual component costs.

⇒ **Final sell product value (15% margin)**

**230 CHF**

# SUSTAINABILITY

## Focusing on 3 R :

**REUSE :**

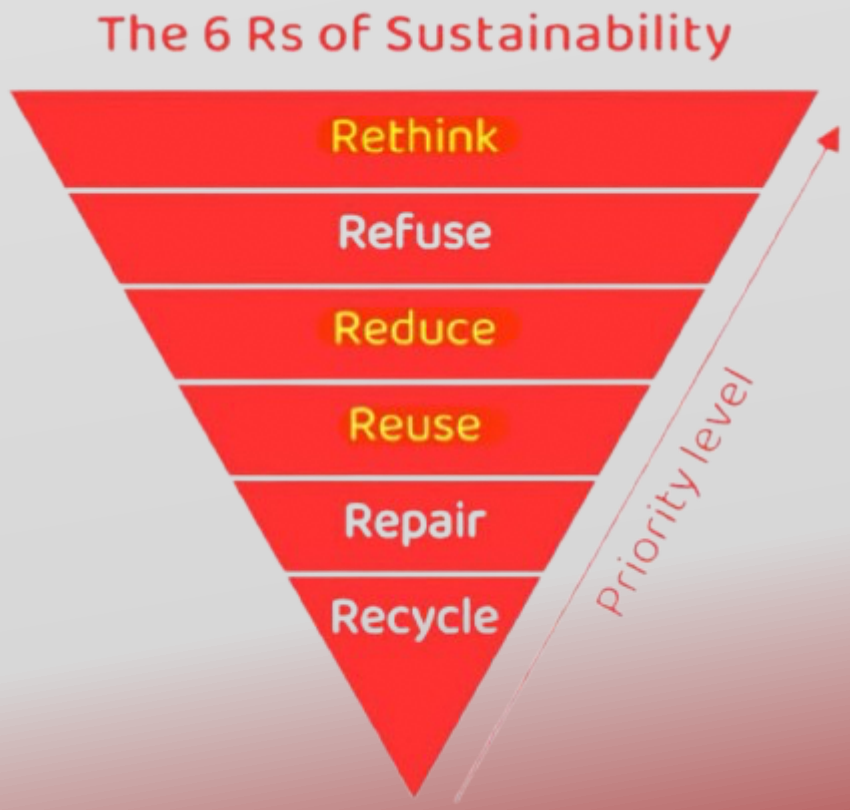
Giving screws a second life

**REDUCE :**

$CO_2$ emission + manual sorting time

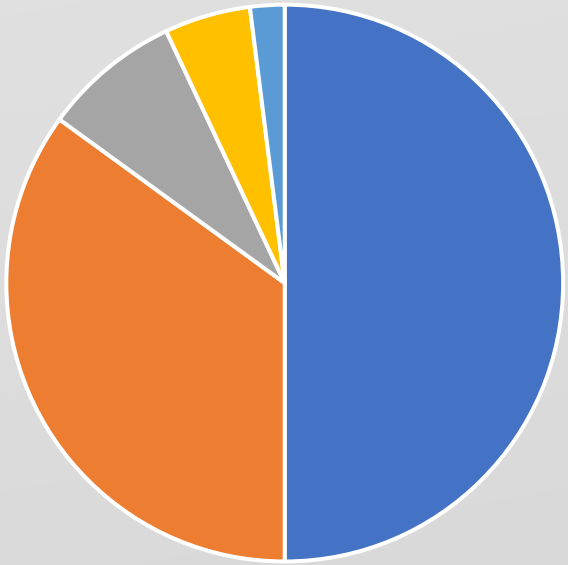
**RETHINK :**

change the way of seeing unusable screws



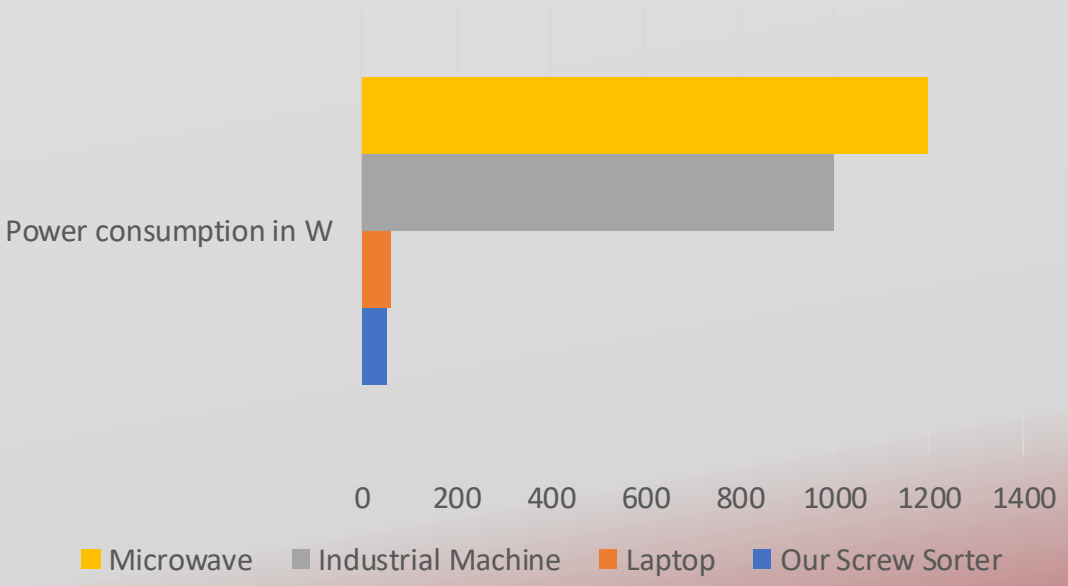
# SUSTAINABILITY

## Material Distribution of the Final Prototype



- MDF : external strcture, base, walls, supports, bins
- PETG: up and down mechanism, some supports
- Aluminium: bearings, 3 axes for the conveyor
- TPU : belt
- Plexiglass : sorting platform

## Instatenous Power Consumption Comparison




- Industrial machines consume around 1000 W (similar to a microwave)
- Our prototype uses only 57 W (as a laptop during normal use and charging).

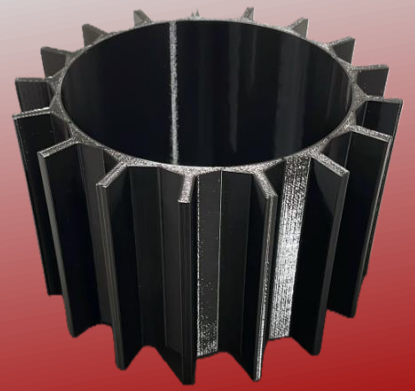
# FUTURE DEVELOPMENT

**1/ Better camera** : improve the accuracy, processing speed and reduce the light dependancy

➔ **Raspberry Pi High Quality Camera**

**2/ 2nd version of the screw separator :** be quicker, more precise and less dependent on it for the synchronisation flux

➔ **New design and new mechanics**

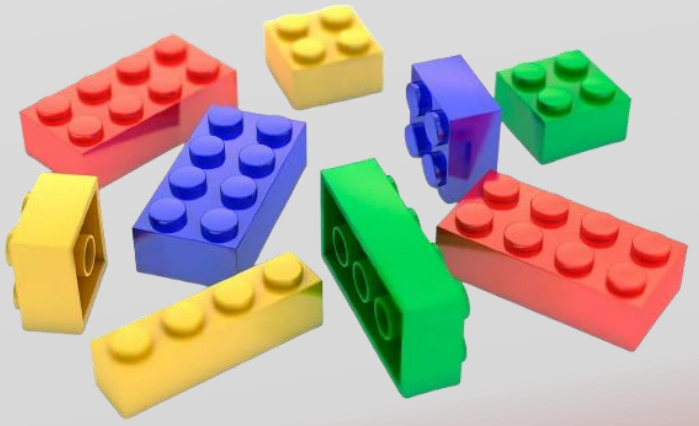


**3/ Improving the printed precision :** reduce the loose of friction on the 3D printed belt

➔ **New design and new mechanics**

# TRANSFER OF OTHER SCENARIOS

Potential to extend the system to sort different objects



➔ With the right choice of extraction methods and a few minor adjustments to the code are needed to handle the new objects.

# Thank You !

Any questions ?

# Sources

[1] : Environmental Product Declaration of screws: https://www.environdec.com/library/epd3969
[2]: Rainbow Screw Sorter : https://www.fastenersmachine.com/products/screw-sorting-machine/